

YIHANG CHEN

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🎓 EDUCATION

Zhejiang University

Sep. 2021 – Jun. 2025 (Expected)

Senior B.Eng. in Computer Science and Technology

Overall GPA: 3.99/4

CS Courses: Fundamentals of Data Structures (98), B/S Architecture Software Design (92), Natural Language Processing (90), Multimedia Technology (96)

Linguistics Courses: Syntax (98), Phonetics (89), Semantics (89), Japanese Pragmatics (96)

👥 INTERNSHIP

Human Language Technology:

• Multimodal AIGC Research Projects

Oct. 2023 – Oct. 2024

Mentor: Prof. Chuang Gan at *University of Massachusetts, Amherst*

Participated in research projects focusing on speech synthesis subsystems.

- Crawled data from the Internet and conducted automatic annotation.
- Trained the speech synthesis subsystems and the baselines on our custom dataset.
- Measured objective metrics on the speech synthesis systems to evaluate their performance.

• Multilingual Text-to-Speech Synthesis

Jul. 2024 – Sept. 2024

Mentor: Prof. Zhizheng Wu at *Chinese University of Hong Kong, Shenzhen*

Participated in a multilingual TTS project based on large language models (LLM) and discrete diffusion.

- Introduced Mamba to the autoregressive model architecture to try to improve inference efficiency.
- Modified the text front-end of the text-to-speech pipeline.
- Tried adapting the pre-trained model to new speakers with parameter-efficient finetuning.

• Multitask Model for Speech Generation

Feb. 2025 – Present

Employer: Tencent AI Lab

We aim to develop speech generation multitask models that, through aligning different modalities of a same data sample, can outperform single-task models on tasks including text-to-speech and voice conversion.

Software Development:

• Digital Humanities Infrastructures Development

Nov. 2024 – Present

Employer: PKU-ByteDance Open Lab for Digital Humanities

We aim to develop user-friendly digital humanities research tools for humanities scholars. The functions include named entity recognition, map visualization, and AI agents that answer questions based on classical texts.

Linguistics:

• Phonetic & Psycholinguistic Data Mining on CHILDES Corpus

Jul. 2023 – Jul. 2024

Mentor: Prof. Yunqi Wang at *Zhejiang University*

Measured the pronunciation difference of polysemous words in different semantic contexts, and investigated its effect on the infants' acquisition of the words.

- Parsed metadata and conversation transcriptions from CHAT files using PyLangAcq.
- Extracted acoustic metrics with tools including MFA, librosa and the Python interface of Praat.
- Investigated the relationship between pronunciation differences and infants' word acquisition through regression analysis.

PREPRINTS AND PAPERS

- Chen, J., Yan, X., **Chen, Y.**, Cen, S., Ma, Q., Zhen, H., ... & Gan, C. (2024). *RapVerse: Coherent Vocals and Whole-Body Motions Generations from Text*. <https://arxiv.org/abs/2405.20336> (CVPR 2025 submitted)
- Chen, J., Fu, Yang, Zeng, A., Wang, Zixin., Cen, Siyuan., Yu, Xueyang., Tanke, J., **Chen, Y.**, ... & Gan, C. (2024). *Orator: LLM-Guided Multi-Shot Speech Video Generation* <https://oratordemo.github.io/> <https://openreview.net/forum?id=Bz6eAiOjrI> (ICLR 2025 under review)

SKILLS

- Python = C/C++ (proficient) > Java = Golang = MATLAB (with project experience)
- Basic machine learning skills with proficiency in **PyTorch**
- Domain knowledge in **natural language processing** (NLP) and **speech processing**
- Domain knowledge in **linguistics**, especially in phonetics and computational linguistics

LANGUAGES

- Proficient in **English** (TOEFL 104)
- Proficient in **Japanese** (JLPT N2)
- Novice at **Arabic** and **Hebrew**, interested in Semitic languages
- Native speaker of **Chinese**

BLOGS

- A course note on ML applied to music audio signal: <https://kaleo996.github.io/eecs/gct634/>
- A course note on music signal processing: <https://kaleo996.github.io/eecs/mpa/>
- A note on Biblical Hebrew: <https://kaleo996.github.io/ling/heb/>
- A course note of generative grammar at Zhejiang University: <https://kaleo996.github.io/ling/syntax/>

HONORS AND AWARDS

2 nd Prize, Zhejiang University Scholarship (top 8%)	Nov. 2022
Zhejiang Provincial Government Scholarship	Nov. 2022
3 rd Prize, Zhejiang University Scholarship (top 20%)	Nov. 2023
3 rd Prize, Zhejiang University Scholarship (top 20%)	Nov. 2024